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| Team name: | Dry Lettuce |
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|                            |                           |                     |
|----------------------------|---------------------------|---------------------|
| Name of the Local Currency | Total Cost Local Currency | Total Cost (\$) USD |
| Australian Dollar AUD      | \$ 3,919.16               | \$ 2,743.41         |

| #  | Component                 | Part name                                                                    | Author                              | Source                                                                        | Quantity | Unit cost (Local Currency) | Unit cost (USD) | Total Cost (Local Currency) | Total Cost (USD) |
|----|---------------------------|------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------|----------|----------------------------|-----------------|-----------------------------|------------------|
| 1  | PCB + assembly/components | Robot PCBs + Front LED PCBs (including IMU, STM32, etc)                      | Toby, William, Landon, Bain, JLCPCB | <a href="http://jlcpcb.com">jlcpcb.com</a>                                    | 5        | \$ 207.00                  | \$ 144.90       | \$ 1,035.00                 | \$ 724.50        |
| 2  | Wheels                    | Pololu Multi-Hub Wheel 2-Pack                                                | Pololu                              | <a href="https://core-electronics.com.au">https://core-electronics.com.au</a> | 4        | \$ 21.30                   | \$ 14.91        | \$ 85.20                    | \$ 59.64         |
| 3  | Motors                    | Pololu 100:1 Metal Gearmotor 20Dx44L mm 12V CB with Extended Motor Shaft     | Pololu                              | <a href="https://core-">https://core-</a>                                     | 4        | \$ 55.80                   | \$ 39.06        | \$ 223.20                   | \$ 156.24        |
| 4  | Servos                    | FT90R - 9g Digital Continuous Rotation Micro Servo                           | Feetech                             | <a href="https://core-">https://core-</a>                                     | 15       | \$ 6.15                    | \$ 4.31         | \$ 92.25                    | \$ 64.58         |
| 5  | Robot base                | Laser cut 4mm black acrylic                                                  | Toby, William, Landon, Bain         | Laser cut                                                                     | 4        | \$ 1.00                    | \$ 0.70         | \$ 4.00                     | \$ 2.80          |
| 6  | Fan                       | Noctua NF-A4x20 40mm 5V 5000RPM PWM Fan(NF-A4x20-5V-PWM)                     | Noctua                              | <a href="https://www.mwave.com.au/">https://www.mwave.com.au/</a>             | 4        | \$ 35.00                   | \$ 24.50        | \$ 140.00                   | \$ 98.00         |
| 7  | Raspberry Pi              | Raspberry Pi 5 Model B 8GB                                                   | Raspberry Pi                        | <a href="https://core-">https://core-</a>                                     | 2        | \$ 285.65                  | \$ 199.96       | \$ 571.30                   | \$ 399.91        |
| 8  | NPU                       | Raspberry Pi AI Hat+ 2                                                       | Raspberry Pi + Hailo                | <a href="https://core-">https://core-</a>                                     | 1        | \$ 289.00                  | \$ 202.30       | \$ 289.00                   | \$ 202.30        |
| 9  | Active CoolerCooler       | Raspberry Pi 5 Active Cooler                                                 | Raspberry Pi                        | <a href="https://core-">https://core-</a>                                     | 1        | \$ 8.80                    | \$ 6.16         | \$ 8.80                     | \$ 6.16          |
| 10 | Cameras                   | Raspberry Pi Camera Module 3 Wide                                            | Raspberry Pi                        | <a href="https://core-">https://core-</a>                                     | 6        | \$ 62.00                   | \$ 43.40        | \$ 372.00                   | \$ 260.40        |
| 11 | Screws                    | Various hex screws (M2, M2.5, M3) + nuts                                     | Multiple                            | Multiple                                                                      | 1        | \$ 50.00                   | \$ 35.00        | \$ 50.00                    | \$ 35.00         |
| 12 | Cables                    | Jumper wires and similar                                                     | Elegoo                              | <a href="https://www.amazon.com.au">https://www.amazon.com.au</a>             | 1        | \$ 10.00                   | \$ 7.00         | \$ 10.00                    | \$ 7.00          |
| 13 | Front LEDs                | White LEDs White 6200 K 70-CRI, XLamp XPGBWLT                                | Cree                                | <a href="https://au.mouser.com/Prod">https://au.mouser.com/Prod</a>           | 16       | \$ 2.03                    | \$ 1.42         | \$ 32.48                    | \$ 22.74         |
| 14 | Down LEDs                 | XL-2835UWC-02                                                                | XINGLIGHT                           | <a href="http://jlcpcb.com">jlcpcb.com</a>                                    | 30       | \$ 0.01                    | \$ 0.01         | \$ 0.30                     | \$ 0.21          |
| 15 | Battery                   | 2x Zeee 3S Lipo Battery 7200mAh 11.1V 120C XT60                              | Zeee                                | <a href="https://www.ebay.com.au/it">https://www.ebay.com.au/it</a>           | 2        | \$ 115.99                  | \$ 81.19        | \$ 231.98                   | \$ 162.39        |
| 16 | Battery alarm             | LiPo Voltage Tester and Alarm                                                | Core Electronics                    | <a href="https://core-electronics.com.au">https://core-electronics.com.au</a> | 3        | \$ 2.40                    | \$ 1.68         | \$ 7.20                     | \$ 5.04          |
| 17 | USB-C cable               | SooPii 100W Right Angle USB C to USB C Cable                                 | SOOPii                              | <a href="https://www.amazon.com.au">https://www.amazon.com.au</a>             | 5        | \$ 13.99                   | \$ 9.79         | \$ 69.95                    | \$ 48.97         |
| 18 | GPIO 40-pin cable         | Raspberry Pi B - GPIO Ribbon Cable (40-pin, 150mm)                           | Sparkfun                            | <a href="https://core-">https://core-</a>                                     | 5        | \$ 9.15                    | \$ 6.41         | \$ 45.75                    | \$ 32.03         |
| 19 | OLED screen               | White I2C OLED display (SSD1306)                                             | Core Electronics                    | <a href="https://core-">https://core-</a>                                     | 5        | \$ 7.20                    | \$ 5.04         | \$ 36.00                    | \$ 25.20         |
| 20 | GPIO extension header     | GPIO Stacking Header for Pi [Extra-long 2x20 Pins]                           | Pi Australia                        | <a href="https://pi4australia.com.au/pr">https://pi4australia.com.au/pr</a>   | 5        | \$ 5.70                    | \$ 3.99         | \$ 28.50                    | \$ 19.95         |
| 21 | JST Connectors            | PH/XH Connector Socket Kit with pre-Crimped Cable Wire                       | elechawk                            | <a href="https://www.amazon.com.au">https://www.amazon.com.au</a>             | 2        | \$ 20.00                   | \$ 14.00        | \$ 40.00                    | \$ 28.00         |
| 22 | Threaded inserts          | 500Pcs Metric Knurled Nuts Assortment Kit Threaded Inserts                   | ZJMCAJY                             | <a href="https://www.amazon.com.au">https://www.amazon.com.au</a>             | 1        | \$ 25.00                   | \$ 17.50        | \$ 25.00                    | \$ 17.50         |
| 23 | Polyfuse                  | Resettable Fuses - PPTC 16V .9A-HD 100A MAX                                  | Bourms                              | <a href="https://au.mouser.com/Prod">https://au.mouser.com/Prod</a>           | 5        | \$ 1.52                    | \$ 1.06         | \$ 7.60                     | \$ 5.32          |
| 24 | Motor Drivers             | MAX14870ETC+T                                                                | Maximum Integrated                  | <a href="http://jlcpcb.com">jlcpcb.com</a>                                    | 20       | \$ 8.09                    | \$ 5.66         | \$ 161.80                   | \$ 113.26        |
| 25 | STM32                     | STM32F407VGT6                                                                | ST Electronics                      | <a href="http://jlcpcb.com">jlcpcb.com</a>                                    | 5        | \$ 8.00                    | \$ 5.60         | \$ 40.00                    | \$ 28.00         |
| 26 | Extra PCB components      | Spare motor drivers, led drivers, current sense resistors, 12v mosfets,      | Various                             | <a href="https://au.mouser.com">https://au.mouser.com</a>                     | 1        | \$ 120.00                  | \$ 84.00        | \$ 120.00                   | \$ 84.00         |
| 27 | Standoffs                 | Brass M2.5 Standoffs for Pi HATs - Black Plated - Pack of 2                  | Adafruit                            | <a href="https://core-">https://core-</a>                                     | 5        | \$ 1.90                    | \$ 1.33         | \$ 9.50                     | \$ 6.65          |
| 28 | 3D printed components     | Sensor/servo/motor mounts, Pi/STM32 enclosure, ball ramp/gate, claw assembly | Toby, William, Landon, Bain         | 3D printed                                                                    | 1        | \$ 50.00                   | \$ 35.00        | \$ 50.00                    | \$ 35.00         |
| 29 | Time of flight sensor     | VL53L1X                                                                      | ST Electronics                      | <a href="https://core-">https://core-</a>                                     | 4        | \$ 20.75                   | \$ 14.53        | \$ 83.00                    | \$ 58.10         |
| 30 | Ultrasonic sensor         | HC-SR04                                                                      | Core Electronics                    | <a href="https://core-">https://core-</a>                                     | 4        | \$ 1.95                    | \$ 1.37         | \$ 7.80                     | \$ 5.46          |
| 31 | BMS                       | Battery Management Li-Ion Li-Phosphate Battery Monitor A 595-BQ7692006PW     | Texas Instruments                   | <a href="https://au.mouser.com/Produ">https://au.mouser.com/Produ</a>         | 5        | \$ 5.27                    | \$ 3.69         | \$ 26.35                    | \$ 18.45         |
| 32 | 12V MOSFET                | MOSFETs 30V N-channel NexFE T Pwr MOSFET A 595-CSD17577Q3AT                  | Texas Instruments                   | <a href="https://au.mouser.com/Prod">https://au.mouser.com/Prod</a>           | 10       | \$ 1.52                    | \$ 1.06         | \$ 15.20                    | \$ 10.64         |

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| Name of the Local Currency | Total Cost Local Currency | Total Cost (\$) U.S.A. Dollars |
|----------------------------|---------------------------|--------------------------------|
| Australian Dollar          | \$ -                      | \$ -                           |

| #  | Name                      | Software - Tool/Library | Description                                                                                                    | Author                                       | Source                                                        | Cost (Local Currency) | Cost (USD) |
|----|---------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------|-----------------------|------------|
| 1  | Python                    | Framework               | High-level, interpreted, general-purpose programming language known for readability and extensive libraries.   | Maintained by the Python Software Foundation | <a href="https://python.org">Python.org</a>                   | \$ -                  | \$ -       |
| 2  | C++                       | Framework               | General-purpose compiled programming language supporting procedural, object-oriented, and generic programming. | Created by Bjarne Stroustrup                 | <a href="https://iso-c++.foundation">ISO C++ Foundation</a>   | \$ -                  | \$ -       |
| 3  | ROS2                      | Framework               | Open-source robotics middleware providing communication, tools, and libraries for robot applications.          | Open Robotics                                | <a href="https://ros2.org">ROS 2</a>                          | \$ -                  | \$ -       |
| 4  | Visual Studio Code        | Tool                    | Free, lightweight, extensible source-code editor with debugging and extension support.                         | Microsoft                                    | <a href="https://visualstudio.code">Visual Studio Code</a>    | \$ -                  | \$ -       |
| 5  | Neovim                    | Tool                    | Hyperextensible Vim-based text editor focused on modern APIs and plugin development.                           | Neovim Project                               | <a href="https://neovim.org">Neovim</a>                       | \$ -                  | \$ -       |
| 6  | Docker                    | Tool                    | Platform for developing, shipping, and running applications in containers.                                     | Docker Inc.                                  | <a href="https://docker.com">Docker</a>                       | \$ -                  | \$ -       |
| 7  | numpy                     | Library                 | Fundamental Python library for numerical computing and multidimensional arrays.                                | NumPy Project                                | <a href="https://numpy.org">NumPy</a>                         | \$ -                  | \$ -       |
| 8  | opencv                    | Library                 | Open-source computer vision and image-processing library.                                                      | OpenCV Team                                  | <a href="https://opencv.org">OpenCV</a>                       | \$ -                  | \$ -       |
| 9  | Fusion360                 | Tool                    | Cloud-based CAD, CAM, CAE, and PCB design platform.                                                            | Autodesk                                     | <a href="https://fusion360.com">Fusion 360</a>                | \$ -                  | \$ -       |
| 10 | Kicad                     | Tool                    | Free and open-source electronic design automation (EDA) suite for PCB design.                                  | KiCad Project                                | <a href="https://kicad.org">KiCad</a>                         | \$ -                  | \$ -       |
| 11 | Github                    | Tool                    | Web-based platform for version control, collaboration, and software development using Git.                     | GitHub Inc.                                  | <a href="https://github.com">GitHub</a>                       | \$ -                  | \$ -       |
| 12 | libcamera                 | Library                 | Open-source camera stack and API for Linux-based camera systems.                                               | libcamera Project                            | <a href="https://libcamera.org">libcamera</a>                 | \$ -                  | \$ -       |
| 13 | BNO08x ros2 driver        | Library                 | ROS 2 package providing integration and sensor data access for BNO08x IMUs.                                    | Balachandra Bhat                             | <a href="https://ros2.bno08x.driver">ROS2 bno08x driver</a>   | \$ -                  | \$ -       |
| 14 | UTM                       | Tool                    | Virtualisation and emulation software for macOS built on QEMU.                                                 | UTM Project                                  | <a href="https://utm.app">UTM</a>                             | \$ -                  | \$ -       |
| 15 | Hailo Software Suite      | Tool/Library            | Software toolkit for compiling, deploying, and optimising AI models on Hailo NPUs.                             | Hailo Technologies                           | <a href="https://hailo.ai">Hailo</a>                          | \$ -                  | \$ -       |
| 16 | YOLO                      | Tool/Library            | Real-time object detection model family that performs object localisation and classification in a single pass. | Maintained by Ultralytics                    | <a href="https://ultralytics.com">Ultralytics YOLO</a>        | \$ -                  | \$ -       |
| 17 | CVAT                      | Tool                    | Open-source web-based annotation tool for creating computer vision datasets.                                   | CVAT.ai                                      | <a href="https://cvat.ai">CVAT</a>                            | \$ -                  | \$ -       |
| 18 | Roboflow                  | Tool                    | Platform for dataset management, annotation, augmentation, and deployment of computer vision models.           | Roboflow Inc.                                | <a href="https://roboflow.com">Roboflow</a>                   | \$ -                  | \$ -       |
| 19 | Custom YOLOv8/Hailo Model | Tool                    | Custom trained vision inference model, made to run on Hailo-10H processor.                                     | Toby                                         | <a href="https://wetlettuce.github.io">Wet Lettuce Github</a> | \$ -                  | \$ -       |